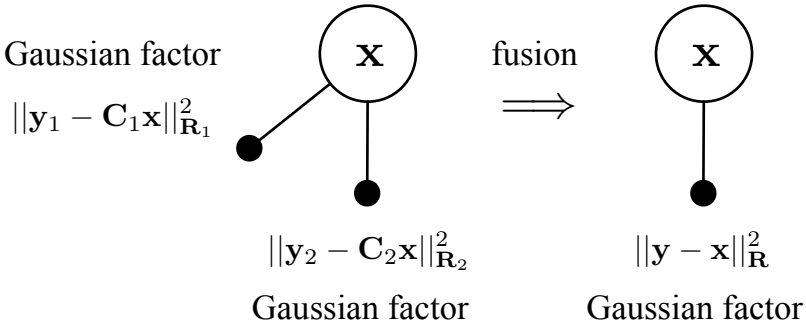

$$\mathbf{R}^{-1} = \mathbf{C}_1^T\mathbf{R}_1^{-1}\mathbf{C}_1 + \mathbf{C}_2^T\mathbf{R}_2^{-1}\mathbf{C}_2$$

$$\mathbf{R}^{-1}\mathbf{y} = \mathbf{C}_1^T\mathbf{R}_1^{-1}\mathbf{y}_1 + \mathbf{C}_2^T\mathbf{R}_2^{-1}\mathbf{y}_2$$